OnTrack: System Features & Specifications

Mobile Application Technology — System Specifications & AI Usage Disclosure

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System Overview: OnTrack is a cross-platform mobile application and web administration portal engineered for On-the-Job Training (OJT) monitoring. It replaces manual Daily Time Records (DTR) by integrating automated hour computation, GPS geolocation verification, selfie proof-of-work, and comprehensive academic coordinator oversight.

PART 1: Client Side (Student Intern Mobile App)

The client side application is built using Ionic 9 and Angular 22 (Standalone Components) paired with Capacitor native plugins. It provides interns with an intuitive interface to log attendance, record daily activities, track required duty hours, and manage academic submissions.

1. Authentication & Onboarding

- **Multi-Step Student Registration:** Guided 3-step registration workflow that captures comprehensive student OJT information.
  - **Step 1 (Academic Details):** Inputs Student ID Number, Full Name, Academic Program, Year Level, and Section selection (BSIT 4-A to 4-K).
  - **Automated Coordinator Pairing:** Selecting an academic section automatically pairs the student with that section's designated faculty coordinator.
  - **Step 2 (OJT Placement):** Records Host Training Establishment (HTE) / Company Name, Assigned Department, Supervisor Name, and Supervisor Contact Details.
  - **Step 3 (Security & Review):** Secure password creation with BCrypt encryption, password visibility toggles, and an instant summary review card before submitting.
- **Student Login & Session Handling:** Secure authentication using Student ID or Institutional Email with hashed password verification. Includes persistent local session storage and Angular Route Guards.

2. Student Dashboard (Home)

- **Profile Header:** Displays user avatar with initials, active duty beacon indicator, student name, student ID, academic section, and assigned company badge.
- **Daily Attendance Card:** Streamlined single-card attendance console with one-tap Clock In / Clock Out button, dynamic duty indicator, and shift start timestamp.

- **Rendered vs. Required Hours:** Real-time counter displaying accumulated rendering hours against the target requirement (e.g., 120.00 / 486 hrs).
- **Visual Progress Ring & Bar:** Live percentage indicator illustrating overall OJT completion progress (0% to 100%).
- **Hours Remaining Metric:** Automatically computed countdown showing exact hours left to complete internship requirements.
- **Target Completion Countdown:** Calculates calendar days remaining until the intern's target completion date.
- **Edit Goal Modal:** Allows students to adjust their required hours target (e.g., 486 or 500 hours) and targeted completion date.
- **Quick Navigation Shortcuts:** 2x2 icon grid providing one-tap access to Time Logs (DTR), Tasks, Reports, and Profile.
- **Recent Activity Feeds:** Instant preview of latest attendance logs and pending tasks directly on the main dashboard feed.

3. Geolocation & Photo Duty Tracking (Time Logs / DTR)

- **GPS Geolocation Verification:** Captures high-accuracy latitude and longitude coordinates upon check-in via @capacitor/geolocation to verify physical workplace presence.
- **Interactive Leaflet Map:** Displays the student's real-time physical GPS location pin on an interactive map inside the clock-in modal.
- **Proof Selfie Camera Integration:** Prompts the student to capture an instant front-facing selfie or duty photo via @capacitor/camera as proof of presence.
- **Tactile Haptic Feedback:** Triggers device vibration via @capacitor/haptics upon successful check-in or check-out.
- **Clock-Out Accomplishment Logging:** Includes an accomplishment summary text area where students log daily duties and tasks finished upon duty completion.
- **Automated Hour Computation:** Accurately computes total hours rendered for the shift based on Time In and Time Out timestamps.
- **DTR History & Timesheet Explorer:** Complete reverse-chronological list of past attendance entries showing date, timestamps, rendered hours, accomplishment logs, and photo verification thumbnails.

4. Task & Deliverable Management (Tasks)

- **Task Status Filtering:** Organizes deliverables into All, Pending, In Progress, and Completed filter tabs.
- **Personal Task Creation:** Interns can create deliverables with custom titles, scope descriptions, and target completion deadlines.
- **Deliverable Submission & Proof Upload:** Allows interns to upload proof files (PDFs, images) or external repository/documentation links to prove task completion.
- **Status Toggling:** One-tap mechanism to update task status from pending to completed.

5. Reports & Profile Management

- **Weekly & Monthly Summaries:** Aggregates weekly rendered hours, daily averages, and attendance consistency.
- **Digital DTR Export:** Prepares structured attendance summary tables ready for printing and academic submission.
- **Profile & Contact Management:** Displays academic details, adviser credentials, and emergency contact details with secure one-tap logout.

PART 2: Admin / Coordinator Side (Academic Faculty Portal)

The coordinator portal provides academic OJT supervisors with real-time oversight tools to monitor student attendance, audit Daily Time Records, inspect GPS coordinates and photo evidence, review student deliverables, and filter cohorts by academic section.

1. Coordinator Authentication & Section Filtering

- **Faculty Authentication:** Dedicated faculty sign-in using Employee ID (e.g., PROF-2026-001) or institutional email with role-based permissions (admin, coordinator, adviser).
- **Interactive Section Switcher:** Header tab dropdown allowing the coordinator to instantly filter the dashboard across sections (BSIT 4-A to 4-K or All Sections).
- **Automated Roster Scoping:** Selecting a section dynamically scopes all KPI analytics, intern lists, and attendance logs to that class.

2. System-Wide Analytics & KPI Metrics

- **Total Monitored Interns:** Live count of registered students currently active in the selected academic section.
- **Active on Duty Counter:** Real-time count of students who are currently clocked in at their company placements.
- **Total Rendered Hours:** Aggregate sum of all internship hours rendered across the student cohort.
- **Task Completion Metrics:** Visual breakdown of pending versus completed student deliverables.

3. Monitored Intern Roster

- **Live Search & Filter:** Filter students in real time by full name, school ID number, or company placement.
- **Duty Status Chips:** Instant filter pills for All, On Duty (live green beacon), and Off Duty (gray indicator).
- **Intern Summary Cards:** Displays student name, student number, company agency, live duty badge, hours remaining, and progress bar (e.g., 245 / 486 hrs · 50.4%).
- **Quick View Action Button:** One-click 'View >' button on every card that opens the student's complete record.

4. Detailed Student Record & Audit View

- **Student Overview Modal:** In-depth modal presenting the student's complete profile, contact details, total hours rendered, remaining hours, and progress percentage.
  - **Tab 1: Complete Attendance History (DTR Audit):** Audit log showing every shift rendered by the student, including date, exact Time In, Time Out, and rendered hours.
  - **Daily Accomplishment Review:** Inspects the daily accomplishment notes written by the student upon clocking out.
  - **GPS Coordinates & Map Inspection:** Verifies exact GPS latitude and longitude captured at the moment of clock-in.
  - **Photo Proof Viewer:** High-resolution viewer allowing coordinators to inspect selfie/duty photos captured via the camera.
  - **Tab 2: Student Tasks & Deliverables:** Displays all deliverables submitted by the student with status indicators and submitted attachments.

5. Student Task Monitoring & Proof Verification Console

- **Task Overview Metrics:** Summary statistics of Total Tasks, Pending Tasks, and Completed Tasks across the section.
- **Deliverables Table:** Interactive table displaying student name, task title, completion date, and file attachment badges.
- **Document & Proof Inspection:** One-click modal to preview and download submitted PDF documents, diagrams, or image proofs.
- **Official Deliverable Verification Stamp:** Allows the coordinator to officially verify and approve submitted internship milestones with a digital verification stamp and timestamp.

6. Real-Time Attendance & Verification Feed

- **Live Duty Feed:** Chronological stream of real-time check-in events across the student cohort.
- **Compliance Audit:** Enables immediate auditing of whether students arrive at their workplaces on time and within assigned geographical boundaries.

PART 3: Feature Comparison Matrix

The table below compares the functional responsibilities between the Student Intern and Academic Coordinator sides of the platform:

Capability / Functional Area	Student Intern (Client Side)	Academic Coordinator (Admin Side)
Authentication & Access	Multi-step registration & student login	Faculty ID login with role-based access

Section Assignment	Selects section (BSIT 4-A to 4-K)	Switches & filters by sections (A to K)
Duty Status Tracking	Single-tap Clock In / Clock Out	Live monitoring of on-duty interns
GPS Verification	Captures device latitude/longitude	Audits coordinates and location tags
Photo Evidence	Captures duty selfie via camera	Inspects photo proof in Student Record
Rendered Hours Computation	Automated daily & remaining hours	Cohort-wide aggregated hours & stats
Accomplishment Logging	Submits daily accomplishment logs	Audits accomplishment notes per shift
Task & Deliverables	Creates tasks and uploads proof files	Monitors deliverables & verifies proofs
Student Record Audit	Views personal metrics & progress	Audits complete DTR history & deliverables
Timesheet & DTR Export	Exports formatted timesheet summary	Inspects digital timesheets across weeks

PART 4: AI Usage Disclosure

1. Identification of AI Tools Used

- **Primary AI Tool:** Google Antigravity / Gemini AI Coding Assistant (DeepMind Advanced Agentic Environment).
- **Deployment Context:** Integrated agentic pair-programming assistant operating directly inside the codebase workspace.
- **Underlying Capabilities:** Context-aware static analysis, real-time TypeScript/PHP code synthesis, terminal automation, and native mobile build synchronization.

2. How the AI Assisted Development

- **Architecture & Scaffolding:** Assisted in structuring the Ionic 9 and Angular 22 standalone component architecture, configuring routes, models, and type-safe HTTP communication with the backend.
- **Device Hardware Integration:** Guided the configuration of Capacitor plugins (@capacitor/camera for selfie capture, @capacitor/geolocation for workplace GPS coordinate recording, and @capacitor/haptics for tactile user feedback).

- **Backend RESTful API & Database Integration:** Constructed secure PHP REST API endpoints utilizing PDO prepared statements to eliminate SQL injection, implemented CORS headers, and authored the normalized MySQL database schema.
- **Native Android Bridge Configuration:** Generated and synchronized native Android platform assets, configured AndroidManifest.xml permissions (Camera, Location, and cleartext HTTP traffic), and prepared Gradle integration for Android Studio.
- **Bug Diagnosis & Query Optimization:** Diagnosed UI change detection locks during registration, applied Angular ChangeDetectorRef updates, and eliminated database latency by introducing targeted composite indexes on high-traffic tables.

3. What the Group Reviewed, Tested, or Modified

- **UI/UX & Mobile Visual Hierarchy Review:** The group reviewed mobile viewport layouts and removed redundant interface clutter—such as repeated company tags and unnecessary 'Off Duty' badges—to maintain a clean and accessible user experience.
- **Academic Workflow & Section Assignment Review:** Reviewed university coordination requirements and structured automated section mapping (BSIT 4-A through 4-K), ensuring newly registered students are automatically paired with their respective academic coordinators.
- **Database Schema & Integrity Verification:** Audited database tables and schema structures in phpMyAdmin to verify referential integrity, foreign key relations, and accurate storage across ontrack_db.
- **Comprehensive Functional Testing:** Conducted end-to-end testing of student registration, duplicate credential validation (handling HTTP 409 conflict states), GPS coordinate capture, selfie camera uploads, and automated daily rendering hour calculation.
- **Multi-Environment Network Testing:** Tested API connectivity across web browsers (localhost), Android Studio Emulators (10.0.2.2), and physical mobile hardware connected via local Wi-Fi.
- **Administrative Scope Modifications:** Refactored the Coordinator Portal to eliminate unnecessary 'add task' buttons, refocusing the coordinator role purely on oversight, DTR auditing, and student deliverable verification.
- **Performance Tuning:** Added composite database indexes (idx_user_date_time and idx_user_id_desc), resolving slow loading times when fetching detailed student attendance histories.